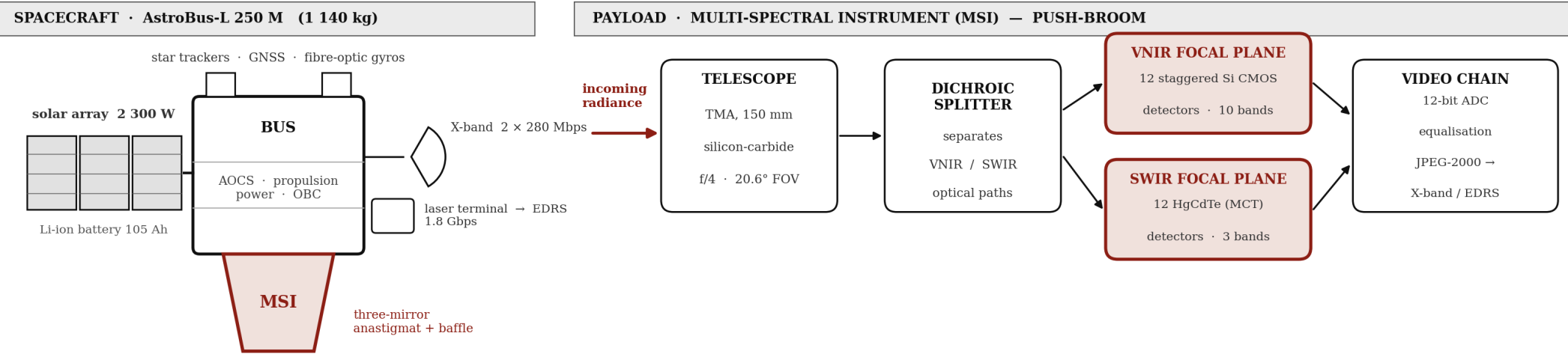


SENTINEL-2 · MSI

Copernicus optical land-monitoring mission — the spatial backbone of this project

786 km SSO · 98.62° · MLST 10:30 descending

5-day revisit (S2A + S2B) · 290 km swath



BUS SUBSYSTEMS

AOCS

3-axis, geodetic pointing

20 m multi-temporal registration

ON-BOARD MEMORY

2.4 Tbit flash

JPEG-2000 wavelet compression ≈ 3:1

PROPULSION

monopropellant

hydrazine, 123 kg

orbit & de-orbit

THERMAL

SWIR FPA passively cooled to ≈ 195 K

SiC optical bench

CALIBRATION, GEOMETRY AND THE GROUND SEGMENT

ON-BOARD CALIBRATION

Sun-illuminated diffuser + shutter

absolute radiometry ≤ 5 %

dark signal from closed shutter

PUSH-BROOM GEOMETRY

no moving scan mirror

one detector line per band

along-track motion builds the image

L1C → L2A (Sen2Cor)

L1C TOA reflectance, ortho, UTM

L2A BOA + Scene Classification

SCL 3/8/9/10/11 masked here

WHAT IT GIVES ME

NBR, dNBR, RBR → severity

NDMI → pre-fire moisture

burnt AREA, the A_i of Seiler-Crutzen

WHY THIS SENSOR

20 m resolves a 309 ha fire into ≈ 7 700 pixels — MODIS gives 3

5-day revisit brackets a 14-day fire

LIMITS I MUST STATE

optical → clouds and smoke block it

no canopy height, no biomass

NBR saturates in dense canopy

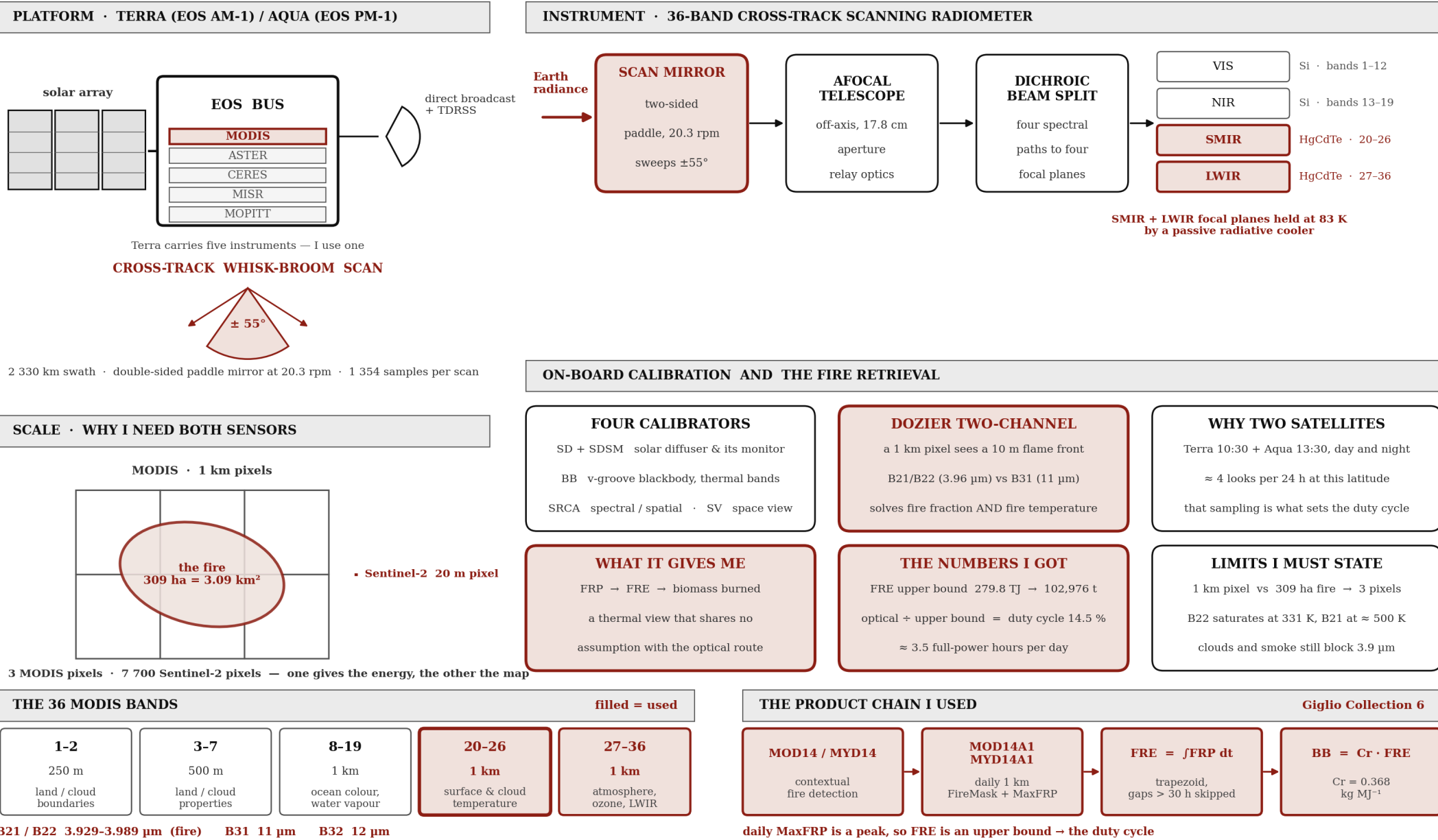
THE 13 MSI BANDS												filled = used in this project
B1 443 nm 60 m	B2 490 nm 10 m	B3 560 nm 10 m	B4 665 nm 10 m	B5 705 nm 20 m	B6 740 nm 20 m	B7 783 nm 20 m	B8 842 nm 10 m NBR	B8A 865 nm 20 m NDMI	B9 945 nm 60 m	B10 1375 nm 60 m	B11 1610 nm 20 m NDMI	B12 2190 nm 20 m NBR

MODIS · TERRA & AQUA

NASA EOS thermal workhorse — the independent energy audit of this project

705 km SSO · Terra 10:30 desc · Aqua 13:30 asc

2 330 km swath · 4 overpasses per day



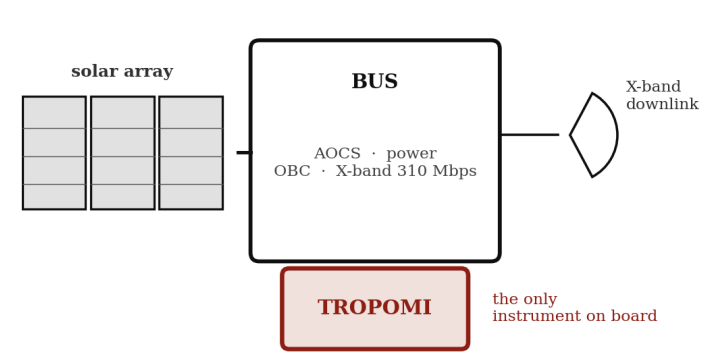
SENTINEL-5P · TROPOMI

Copernicus atmospheric mission — the methane observability test of this project

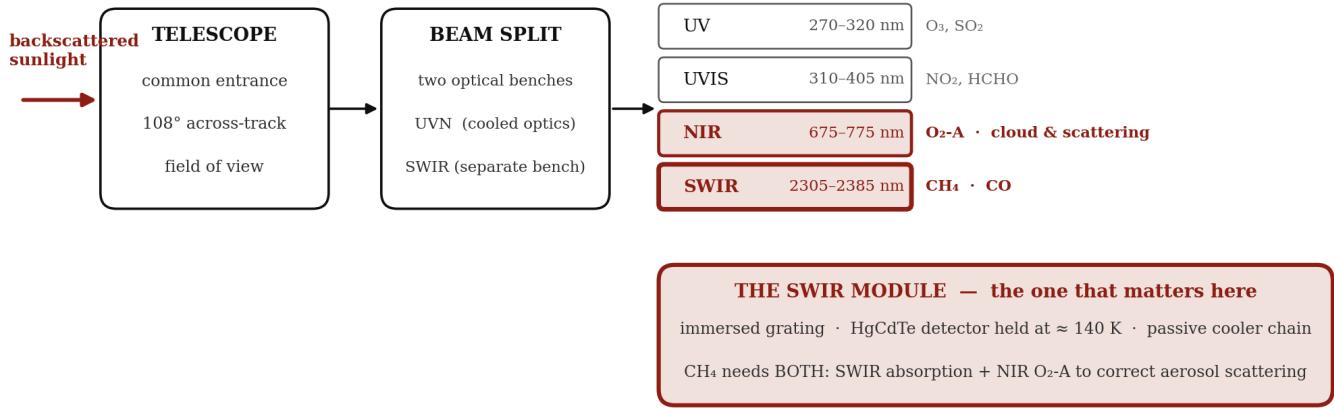
824 km SSO · 98.74° · LTAN 13:30 ascending

2 600 km swath · daily global coverage

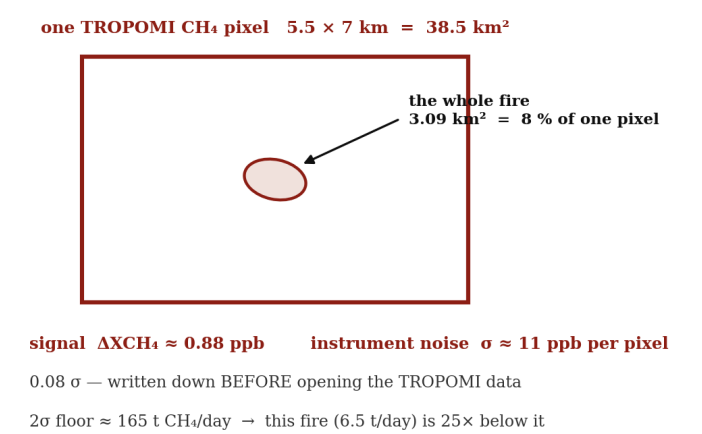
SPACECRAFT · ASTROBUS-L 250 M (≈ 980 kg, single payload)



PAYLOAD · TROPOMI — PUSH-BROOM IMAGING GRATING SPECTROMETER



THE GROUND PIXEL · WHY THIS FIRE IS INVISIBLE



RETRIEVAL AND WHAT I DID WITH IT

RemoTeC FULL-PHYSICS

fits CH₄, H₂O, aerosol size & height

jointly across SWIR + NIR windows

output XCH₄ in ppb, dry-air column

WHAT ppb ACTUALLY MEANS

1 ppb of 1 900 ppb background

= reading a change of 0.05 % —

like 1 grey level in a 12-bit camera

L2_CH4__ · QUALITY

qa_value ≥ 0.5 keeps clear-sky pixels

bias-corrected column mixing ratio

0 valid CH₄ days over the site (CO: 33)

HOW I USED IT

IME mass balance run FORWARDS:

τ = L/U, IME = Q·τ, ΔXCH₄ from column

a prediction, not a retrieval

THE HONEST RESULT

undetectable in all 27 plume geometries

floor 27 – 1,098 t CH₄/day

best case only 0.47 σ

WHAT WOULD SEE IT

GHGSat 25–50 m, but tasked, not free

EnMAP / PRISMA 30 m imaging spectro.

CO2M and MethaneSAT next generation

THE FOUR TROPOMI SPECTROMETERS

UV 270–320 nm O₃, SO₂	UVIS 310–405 nm NO₂, HCHO	NIR 675–775 nm O₂-A · cloud & scattering	SWIR 2305–2385 nm CH₄ · CO
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